

High-Pressure Infrared Spectroscopy System

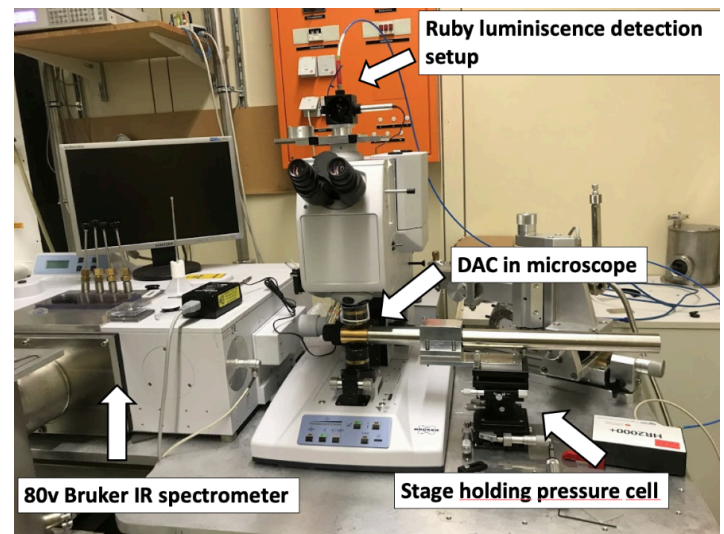
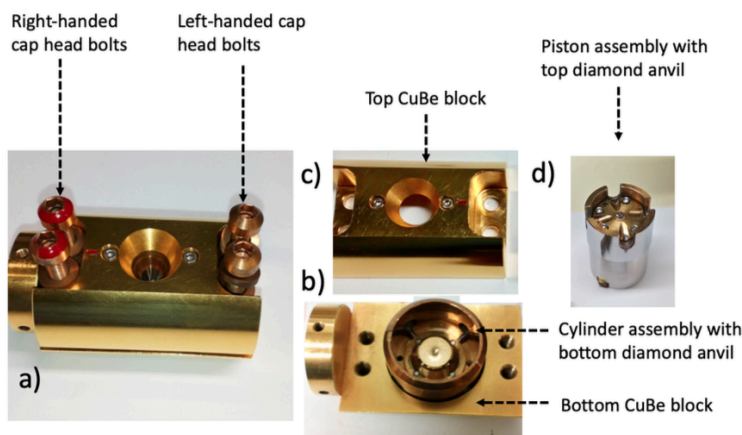
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Project Overview

Designed and implemented experimental setups for infrared-terahertz spectroscopy measurements under high pressure conditions to investigate electronic and optical properties of materials. The system integrates spectroscopy instrumentation, precision mechanics, pressure cells and optical and electronic measurement techniques to perform measurements under controlled pressure environments, including sensor integration, system calibration, signal acquisition and diagnostic analysis of complex measurement systems.

System Components

infrared spectrometer, infrared microscope, optical interferometer, broadband infrared radiation source, beam splitters and optical mirrors, electronic and optical detectors, photodiode sensors, spectroscopy data acquisition systems, optical alignment lasers, diamond anvil pressure cell, precision translation and alignment monitored stages, adjustable optical mounts and positioning platforms, detector electronics and signal amplifiers, precision and calibration instrumentation.



System Architecture

Infrared radiation source → interferometer → beam splitter → optical microscope → sample in diamond anvil pressure cell → reflected signal → detector → signal amplifier → automated spectroscopy data acquisition system → analysis software.

Technical Work Performed

- Installation and alignment of optical, mechanical and electronic components as well as the measurement instrumentation
- Configuration and operation of infrared spectroscopy systems
- Integration of high pressure measurement cells with microscopy setups
- Calibration of detectors, optics and pressure measurement components
- Signal acquisition and interpretation of spectroscopic data
- Troubleshooting measurement instability and alignment errors
- Performed systematic diagnostics to identify signal noise sources, alignment errors and measurement instabilities in the spectroscopy system
- Mechanical assembly and configuration of experimental setups

Scientific Output

Results from this experimental work contributed to peer reviewed publications:

- <https://journals.aps.org/prl/abstract/10.1103/PhysRevLett.124.136402>
- <https://journals.aps.org/prb/abstract/10.1103/PhysRevB.101.174104>

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